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/* =====
/* OPERATORS Section
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/* Types of Operators in JS
// Assignment operators
// Arithmetic operators
// String operators
// Comparison operators
// Logical operators
// Unary operator
// Conditional (ternary) operator

/* =====
/* 1: Assignment operators
/* =====
// Assignment operators in programming are symbols used to assign values to variables. They take

// examples
// var myFavNum = 12;
// Assigns the value 12 to the variable myFavNum
// console.log(myFavNum)

/* =====
/* 2: Arithmetic operators
/* =====

// Arithmetic operators in programming perform basic mathematical operations on variables or values.

// Addition (+): Adds two values or variables.
// Example:
// var x = 5;
// var y = 10;
// var sum = x + y;
// console.log(sum);

// Subtraction (-): Subtracts the right operand from the left operand.
// Example:
// var a = 10;
// var b = 7;
// var difference = a - b;
// console.log(difference);

// Multiplication (*): Multiplies two values or variables.
// Example:
// var p = 4;
// var q = 6;
// var product = p * q;
// console.log(product);

// Division (/): Divides the left operand by the right operand.
// Example:
// var m = 15;
// var n = 3;
// var quotient = m / n;
// console.log(quotient);

// Modulus (%): Returns the remainder when the left operand is divided by the right operand.
// Example:

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// var c = 17;  
// var d = 5;  
// var remainder = c % d;  
// console.log(remainder);
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```
// Special case  
// var result = "hello" / 2;  
// console.log(result);
```

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/* =====  
/* 3: String Operators  
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// There are a few ways to concatenate strings in JavaScript. The most common way is to use the +

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// var str1 = "Hello";  
// var str2 = "World ";  
// var str3 = str1 + str2;  
// console.log(str3);
```

```
// special case  
// var a = "5";  
// var b = 3;  
// var sum = a + b;  
// var sum = parseInt(a) + b;  
// console.log(sum);
```

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/* =====  
/* 4: comparison operators  
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// Comparison operators in JavaScript are used to compare values and return a Boolean result (true or false)

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// Equal (==): Checks if two values are equal.  
// console.log(5 == "5");
```

```
// Strict Equal (===):  
// Checks if two values are equal with also check their datatype.  
// console.log(5 === "5");
```

```
// Not Equal (!=):  
// Checks if two values are not equal.  
// console.log(5 != 5);
```

```
// Greater Than (>):  
// Checks if the value on the left is greater than the value on the right.  
// console.log(5 > 2);
```

```
// Less Than (<):  
// Checks if the value on the left is less than the value on the right.  
// console.log(5 < 10);
```

```
// Greater Than or Equal To (>=):  
// Checks if the value on the left is greater than or equal to the value on the right.  
// console.log(10 >= 10);
```

```
// Less Than or Equal To (<=):  
// Checks if the value on the left is less than or equal to the value on the right.  
// console.log(5 <= 10);
```

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/* =====  
/* 5: Logical operators in JavaScript  
/* =====
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/* There are three main logical operators: && (logical AND), || (logical OR), and ! (logical NOT).

// Logical AND (&&): Returns true if both operands are true, otherwise, it returns false.

// Example:

// var x = 5;

// var y = 10;

// console.log(x > 0 && y < 0);

// Logical OR (||): Returns true if at least one of the operands is true, otherwise, it returns false.

// Example:

// var a = 15;

// var b = 0;

// console.log(a > 10 || b > 10);

// Logical NOT (!):

//? Returns true if the operand is false, and false if the operand is true.

// Example:

// var isOpen = false;

// console.log(!isOpen);

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/* =====  
/* 6: Unary operator  
/* =====
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//? Prefix Increment (++x) and Prefix Decrement (--x): In prefix form, the value of the operand is first

// var x = 5;

// var y = --x;

// console.log(y);

// console.log(x);

//? Postfix Increment (x++) and Postfix Decrement (x--): In postfix form, the value of the operand is

// var x = 5;

// var y = x++;

// console.log(y);

// console.log(x);

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/* =====  
/* 7: Conditional (ternary) operator  
/* =====
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//? syntax: condition ? expressionIfTrue : expressionIfFalse;

// ! write a program to check if the candidates isEligibleForDrive or not? Age must be equal to or greater

// var age = 19;

// var result = age >= 18 ? "Yes" : "No";

// console.log(result);